

DDTA research work plan after documentation-layer closure

WORK PLAN - REVISION 10

Status: active BA2 execution plan after BA2-T2 operator/role/modifier vocabulary pressure testing.

Supersedes: Revision 9 only for forward execution state; R1-R9 remain historical research records.

1. Fixed prior state

- Chapters 2-4: CLOSED / FINAL for current thesis scope.
- Documentation layer: CLOSED.
- BA0-R systems-modeling prior-art research: CLOSED.
- BA0 responsibility/non-goals: CLOSED.
- BA1 minimal BAE identity ontology: CLOSED.
- **BAReferent** and **BAProposition**: ACCEPTED first-class semantic identity families.
- ThreatForge is a case study/tooling instrument, not DDTA semantic authority.

2. Closed BA1 identity boundary

BA1 accepts exactly:

BAReferent

BAProposition

BA2 must not introduce **Action, Relation, Participation, Role, Behavior, Information, Store, Contract, Boundary** or **State** as new first-class identity families unless a concrete BA1 reopen criterion is met.

3. BA2 objective

BA2 defines the smallest stable methodology-neutral proposition/relation/action semantics over the BA1 identities while preserving:

```
semantic identity family
    !=
semantic operator / participation role / modifier
    !=
lexical label / authoring synonym
```

Base Analysis must support mechanical selection and projection without forcing consumers to reinterpret raw prose or import threat-method categories.

4. BA2-T1 - proposition structural lower bound

Status: COMPLETED / PROVISIONAL CANDIDATE.

BA2-T1 rejects pure binary SPO as the universal proposition shape and retains explicit n-ary role-bound participation with a stable semantic operator and scoped modifier capability.

BAProposition

```
| - semanticOperator    1
| - participation      1..*
|   | - role           1
|   ` - term           1
` - scopedModifier     0..*
```

5. BA2-T2 - operator, role and modifier vocabulary pressure test

Status: COMPLETED / PROVISIONAL PASS WITH VOCABULARY-ARCHITECTURE REFINEMENT.

BA2-T2 pressure-tested source-predicate copying, operator grouping, role scoping, modifier boundaries, polarity, classification and the bounded method consumer.

Result

Stable semantic operator key	REQUIRED
Source predicate list as BA vocabulary	REJECTED
Lexical label as semantic identity	REJECTED
Operator-family facet	PROVISIONAL ORGANIZING CANDIDATE
Seed operator key set	PROVISIONAL CANDIDATE
Reusable role concepts	SUPPORTED
Operator-scoped role compatibility/cardinality	REQUIRED
Global context-free role contract	REJECTED
Untyped/free-text modifier bag	REJECTED
Normalized local modifier capability	SUPPORTED / CANDIDATE
Explicit polarity	STRONG CANDIDATE
Modifier promotion rule	REQUIRED
Classification-as-BAProposition	STRONG CANDIDATE
Exact semantic-kind vocabulary	OPEN
Third BA1 identity family	NOT FORCED

The active vocabulary candidate is BA2_SEMANTIC_VOCABULARY_CANDIDATE_R1.md.

Semantic boundaries retained

BA2-T2 explicitly preserves:

correlation	!= generic reference
service consumption	!= ownership/management transfer
provider raw state	!= governed project state
request acceptance	!= physical handoff
realization	!= generic local modifier when reusable
referent classification	!= participation role

Modifier promotion rule

A modifier remains embedded only when it is local to one proposition/binding, has no new independent participant set, needs no independent assertion provenance/review identity, and is not reused as project meaning.

Otherwise model the meaning through explicit BAreferent/BAProposition semantics rather than a modifier mini-language.

6. BA2-T3 - cross-corpus regression and semantic-key/lexical separation

NEXT / NOT EXECUTED BY THIS PLAN REVISION.

BA2-T3 must replay the BA2-T2 candidate systematically against both current corpora rather than inventing more vocabulary for reassurance.

It must test:

1. whether every seed operator key is necessary or whether keys can merge without semantic loss;
2. whether any reviewed fact requires a missing method-neutral key or role distinction;

3. whether operator-scoped role contracts can assign compatible minimum/maximum cardinalities consistently;
4. whether the same semantic key remains stable across different source verbs and whether lexical synonyms can be separated cleanly from semantic identity;
5. whether explicit polarity avoids negative operator proliferation without weakening constraint semantics;
6. whether the modifier promotion rule keeps correlation, realization, responsibility and reusable constraints out of a generic modifier bag;
7. whether classification-as-proposition supplies the reusable semantic-kind distinctions needed by human/method consumers without reopening BA1;
8. whether the bounded STRIDE-oriented projection remains constructible without DFD/STRIDE leakage.

Exit condition

Produce a regression disposition for each candidate key/role/modifier rule and identify the smallest unresolved set.

If the candidate survives but closure criteria still require a distinct review, schedule that review. Do not close BA2 merely because the current examples can be encoded, and do not invent additional microsteps without a concrete unresolved falsification target.

7. BA3 - derivation, provenance and authority

Only after BA2 closes, BA3 must close:

- document-to-BASemantic/BASemantic derivation rules;
- source locators;
- grounded, derived and diagnostic/unresolved materialization;
- identity/equivalence/lifecycle rules;
- acceptance/rejection/stale semantics;
- source-to-analysis and analysis-to-source traceability.

Governed documentation remains project authority.

8. BA4 - projections

Define derived human and method projections from the same accepted BA semantics. No view becomes source authority.

9. BA5 - lexical vocabulary and optional assistance

Only after semantic operator/role contracts are closed, define canonical display labels, authoring synonyms and optional assistance. NLP/LLM proposals remain candidate-producing support and cannot establish project semantics silently.

10. BA6 - Base Analysis closure and regression

Regress the complete Base Analysis design against closed corpora and at least one structurally different holdout. BA6 closes the complete Base Analysis design.

11. Analysis envelope remains downstream

Only after Base Analysis closure:

- A1 - AnalysisRecord / execution envelope;
- A2 - common reviewed Finding boundary;
- A3 - change/provenance integration;
- formal methodology overlays and STRIDE/STRIDE-AI evaluations.

The earlier STRIDE transfer probe remains only a bounded consumer probe.

12. Next authorized microstep

After the BA2-T2 package is reviewed, committed and remotely verified, execute only:

BA2-T3 - cross-corpus regression of the operator/role/modifier candidate and semantic-key/lexical separation.

Do not start BA3, formal STRIDE overlay design, Common Finding schema or implementation work.